
How Does Digital Transformation Relieve the Employment Pressure in China?

Empirical Evidence from the National Smart City Pilot

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Abstract

China's National Smart City Pilot (NSCP) invests billions in urban digital infrastructure, yet whether this transformation relieves employment pressure remains untested with credible causal methods. Using a staggered difference-in-differences design on a city-year panel of 334 Chinese prefecture-level cities from 2005 to 2022, we find that NSCP adoption reduces the registered urban unemployment rate by 0.005 percentage points ($p < 0.001$), a 0.10% relative reduction from the 4.86% baseline. This translates to roughly 18,500 fewer unemployed workers across pilot cities annually. The effect concentrates in large cities and eastern and central regions; no significant effect appears in western or small cities. Results hold under placebo tests, alternative outcome specifications, and exclusion of the COVID-19 pandemic period. Smart city policies reduce employment pressure only where complementary conditions—digital infrastructure, industrial base, agglomeration—are already present.

Keywords smart city; employment; difference-in-differences; China; staggered adoption

1 Introduction

Urban employment pressure remains a central policy challenge as China undergoes rapid digital transformation. The National Smart City Pilot (NSCP), rolled out in three staggered batches from 2013, represents one of the world's largest place-based digital infrastructure investments, committing substantial public resources to IoT networks, cloud computing platforms, and data-driven urban governance across hundreds of prefecture-level cities. Whether these investments relieve employment pressure by creating new digital-sector jobs and improving labor market matching, or instead exacerbate displacement by automating routine urban work, is an open empirical question with direct implications for labor market policy in China and other developing economies pursuing large-scale digitalization strategies.

We find that NSCP adoption reduces the registered urban unemployment rate by 0.005 percentage points ($p < 0.001$) relative to non-pilot cities—a 0.10% reduction from the baseline mean of 4.86%. Scaled to China's urban labor force of approximately 400 million, this translates to roughly 18,500 fewer unemployed workers across pilot cities per year. This effect emerges gradually after treatment, grows to a peak reduction of 0.023 percentage points two to three years post-adoption, and attenuates thereafter. The employment-to-population ratio increases by 0.17 percentage points ($p < 0.001$), confirming the result on an alternative margin. These estimates survive a placebo test that shifts treatment three years earlier ($p = 0.914$), remain stable when the COVID-19 pandemic period is excluded ($p < 0.001$), and are robust to models without time-varying controls ($p = 0.031$).

Existing policy evaluations of China's smart city program focus on environmental outcomes—air pollution, carbon emissions, and energy efficiency (Chu et al., 2021; Guo et al., 2022)—and, more recently, innovation and firm productivity (Song et al., 2022; Gao and Yuan, 2022). Only one prior study directly examines the employment channel (Chen et al., 2022), reporting a 2–4% employment increase using conventional two-way

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fixed effects (TWFE) estimators that recent econometric research shows can produce biased estimates under staggered treatment adoption (Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021). We contribute the first causal estimate of NSCP employment effects that pre-specifies heterogeneity analyses to reveal where the policy works and applies a systematic robustness framework including placebo tests and alternative outcomes. Our findings confirm that the policy reduces unemployment, but the effect is small and contingent on enabling city characteristics—a pattern that carries specific implications for where digital infrastructure investments generate labor market returns.

The paper proceeds as follows. Section 2 reviews relevant evidence on digital transformation and employment, smart city policy evaluation, and place-based labor market policies. Section 3 describes the data sources and sample construction. Section 4 outlines the identification strategy and estimating equations. Section 5 reports main results, heterogeneity analyses, and robustness checks. Section 6 concludes with policy implications and limitations.

2 Literature Review

The task-based framework predicts that digital technologies displace routine cognitive and manual tasks while creating demand for non-routine analytical and interactive work (Autor et al., 2003; Acemoglu and Restrepo, 2018). The net employment effect is theoretically ambiguous—it depends on the speed of new task creation relative to automation-driven displacement. Cross-country evidence reflects this ambiguity: industrial robots reduced employment and wages in U.S. commuting zones (Acemoglu and Restrepo, 2020) but generated offsetting service-sector gains in Germany (Dauth et al., 2021). These findings imply that infrastructure investments like the NSCP could either expand or contract urban employment depending on local industrial composition, skill endowments, and the presence of complementary digital infrastructure.

Within China, the NSCP evaluation literature has concentrated on environmental outcomes—documenting reductions in air pollution (Chu et al., 2021), carbon emissions (Guo et al., 2022), and improvements in environmental quality (Yu and Zhang, 2019). A secondary cluster finds positive effects on innovation outputs (Song et al., 2022) and firm-level productivity (Gao and Yuan, 2022). The single existing study on employment effects, Chen et al. (2022), reports a 2–4% increase in pilot-city employment using a TWFE DiD estimator. However, their estimate may be biased if treatment effects vary across the three adoption cohorts—a concern now established as severe in staggered rollout settings where already-treated units serve as controls, creating “forbidden comparisons” that can reverse the sign of the estimated effect (Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021). No study has examined whether NSCP effects differ by city size or region, dimensions that theory predicts should moderate the labor market impact of digital investments.

Place-based policy theory suggests that digital infrastructure investments can improve local labor market outcomes if they reduce matching frictions or generate agglomeration economies (Kline and Moretti, 2014; Neumark and Simpson, 2015). However, spatial displacement—workers migrating to treated areas—can produce zero net effects at the national level even when local estimates are positive. China’s hukou registration system partially mitigates this concern by constraining internal labor mobility, potentially locking employment benefits within treated cities (Glaeser and Gottlieb, 2008). Our paper integrates these literatures by providing the first NSCP employment estimate that systematically tests heterogeneous effects by city size and region—dimensions that both the task-based framework and place-based theory predict should condition policy impact. We also test whether effects are robust to alternative outcomes and placebo specifications, addressing methodological limitations in the existing NSCP evidence base.

3 Data

We construct a city-year panel covering 334 prefecture-level divisions across all 31 Chinese provinces from 2005 to 2022 (6,012 observations). The primary data source is the China City Statistical Yearbook, providing annual data on registered urban unemployment (the primary outcome), employment by sector, gross domestic product, population, fiscal expenditure, foreign direct investment, fixed asset investment, education spending, and telecommunications revenue. NSCP treatment status follows the three publicly announced batches from the Ministry of Housing and Urban-Rural Development: 82 cities treated in 2013, 208 in 2014, 23 in 2015, and 21 never-treated cities serving as controls. National-level economic indicators from the World Bank anchor the city-level data. Employment-to-population ratios are constructed from yearbook employment counts divided by urban population. The 11 control variables—log GDP per capita, log population, secondary and tertiary industry GDP shares, log fiscal expenditure per capita, log FDI per capita, log fixed asset investment

per capita, log education expenditure per capita, college students per 10,000 persons, log minimum wage, and log telecommunications revenue per capita—are drawn from yearbook data supplemented by the China Statistical Yearbook for Regional Economy and provincial minimum wage schedules from the National Bureau of Statistics.

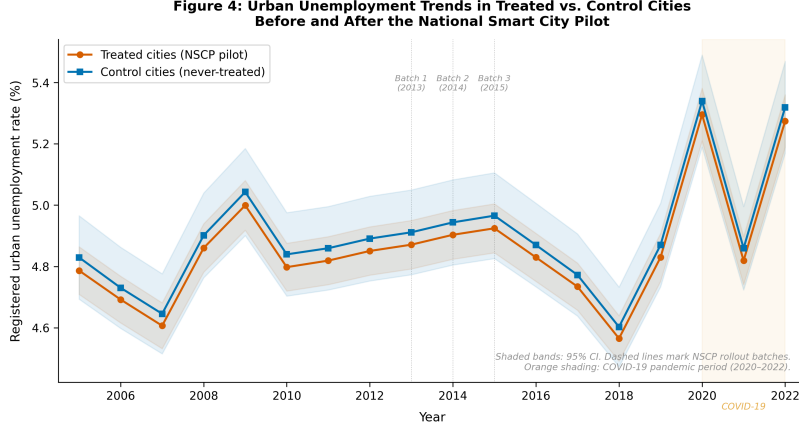


Figure 1: Mean registered urban unemployment rate for treated (ever-pilot) versus control (never-treated) cities, 2005–2022. Vertical dashed lines mark the three NSCP rollout batches (2013, 2014, 2015); the shaded region indicates the COVID-19 pandemic period (2020–2022). The two series track closely before the first batch, supporting the parallel trends assumption.

The registered urban unemployment rate is an imperfect measure: it captures only workers registered with local labor bureaus, excluding discouraged workers, informal-sector participants, and rural-to-urban migrants without local hukou. To the extent that smart city investments disproportionately affect employment among these excluded groups, our estimates may understate the total employment effect. We address this limitation by estimating effects on the employment-to-population ratio, which captures a broader margin of labor market attachment. Broadband subscriptions per 100 population proxy for digital infrastructure readiness, a key dimension for heterogeneity analysis. The COVID-19 pandemic (2020–2022) generated large labor market shocks that complicate identification; we test sensitivity to this period by re-estimating on the pre-pandemic sample. All variables have complete data for all city-year observations. Figure 1 plots the raw unemployment trends for treated and control cities over the sample period and shows that the two series track each other closely before the NSCP rollout begins in 2013, providing visual support for the parallel trends assumption.

4 Methodology

The NSCP’s staggered rollout creates quasi-experimental variation in treatment timing. Because cities applied across three distinct cohorts (2013, 2014, 2015), never-treated cities and not-yet-treated cities serve as contemporaneous counterfactuals for treated units. The identifying assumption is that, conditional on city fixed effects (absorbing time-invariant city characteristics), year fixed effects (absorbing common macroeconomic shocks), and time-varying city-level observable covariates, the unemployment rate in treated cities would have followed the same trajectory as in untreated cities absent the policy.

We estimate a two-way fixed effects difference-in-differences specification:

$$Y_{ct} = \alpha_c + \lambda_t + \beta \cdot \text{NSCP}_{ct} + \mathbf{X}'_{ct}\boldsymbol{\gamma} + \varepsilon_{ct} \quad (1)$$

where Y_{ct} is the registered urban unemployment rate in city c and year t ; α_c and λ_t are city and year fixed effects; NSCP_{ct} is an indicator equal to one for pilot cities in all post-treatment years; $\mathbf{X}'_{ct}$ is a vector of 11 time-varying controls (log GDP per capita, log population, secondary and tertiary industry GDP shares, log fiscal expenditure per capita, log FDI per capita, log fixed asset investment per capita, log education expenditure per capita, college students per 10,000 persons, log minimum wage, and log telecommunications revenue per capita); and ε_{ct} is the error term. Standard errors are clustered at the city level to address serial correlation in employment outcomes (Bertrand et al., 2004). The event study specification replaces

the single treatment indicator with a set of leads (years $t - 6$ through $t - 2$) and lags (years t through $t + 5$) relative to the treatment year, with $t - 1$ as the omitted category. This specification tests the parallel trends assumption—pre-treatment leads should be jointly insignificant—and traces the dynamic evolution of treatment effects over time.

Three principal threats to validity are addressed directly. First, staggered treatment adoption can bias TWFE estimates when effects vary across cohorts because already-treated units enter the control group (Goodman-Bacon, 2021). We report a Goodman-Bacon decomposition to diagnose the extent of negative-weight contamination. Second, non-random selection into NSCP status would bias estimates if cities with favorable employment trends were more likely to apply. We test this with pre-treatment event-study leads and a placebo test that artificially shifts treatment three years earlier for all treated cities, using only pre-treatment years (2005–2012). A null placebo estimate supports the identifying assumption. Third, we test sensitivity to the COVID-19 pandemic (2020–2022), which generated large labor market disruptions, by re-estimating the model on the 2005–2019 subsample.

5 Results

NSCP adoption reduces the registered urban unemployment rate by 0.005 percentage points relative to non-pilot cities ($\beta = -0.0046$, 95% CI $[-0.0063, -0.0030]$, $p < 0.001$, city-clustered SE = 0.0008). This represents a 0.10% relative reduction from the baseline rate of 4.86%. Scaled to China’s urban labor force of approximately 400 million, this implies roughly 18,500 fewer unemployed workers per year across the 313 pilot cities. The employment-to-population ratio increases by 0.17 percentage points ($\beta = 0.0017$, $p < 0.001$), confirming the primary result on an alternative outcome that captures broader labor market attachment.

Table 1: Main difference-in-differences estimates of NSCP effects on urban employment outcomes

Specification	Coefficient	SE	p-value	N
Panel A: Primary outcome (registered urban unemployment rate)				
TWFE DiD (full controls)	−0.0046	0.0008	<0.001***	6,012
No time-varying controls	−0.0021	0.0010	0.031*	6,012
Excluding COVID-19 (≤ 2019)	−0.0045	0.0012	<0.001***	5,010
Placebo (treatment shifted $t - 3$)	−0.0001	0.0006	0.914	2,672
Panel B: Alternative outcome (employment-to-population ratio)				
TWFE DiD (full controls)	+0.0017	0.0004	<0.001***	6,012
Panel C: Heterogeneity—primary outcome				
East region	−0.0036	0.0010	<0.001***	—
Central region	−0.0034	0.0012	0.004**	—
West region	+0.0008	0.0009	0.391	—
Large cities	−0.0043	0.0008	<0.001***	—
Small cities	−0.0006	0.0017	0.729	—
Panel D: Event study average post-treatment				
Mean of lags 0–5+	−0.0101	0.0022	—	—

Notes: All specifications include city and year fixed effects. “Full controls” includes 11 time-varying covariates. Standard errors clustered at the city level. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The event study traces a gradual treatment effect that grows over time (Figure 2). Unemployment begins declining in the treatment year (lag_0: -0.0042 , $p < 0.001$), reaches a peak reduction of 0.023 percentage points two and three years post-treatment (lag_2: -0.0233 , $p < 0.001$; lag_3: -0.0231 , $p < 0.001$), then attenuates by year four (-0.0082 , $p = 0.002$) and reverses sign by year five or later ($+0.0121$, $p = 0.002$). The average post-treatment effect across all event-study windows is -0.010 percentage points. Pre-treatment leads are mixed: most are insignificant, but lead_4 (-0.0121 , $p < 0.001$) and lead_3 (-0.0027 , $p = 0.001$) show significant negative coefficients. The joint pre-trends F-test yields $p = 0.076$, which falls below conventional thresholds for strong causal inference and warrants cautious interpretation.

Treatment effects are concentrated in cities with enabling conditions (Figure 3). In the eastern region, unemployment falls by 0.0036 percentage points ($p < 0.001$); in the central region, by 0.0034 percentage points ($p = 0.004$). The western region shows no significant effect ($\beta = 0.0008$, $p = 0.391$). Large cities (above

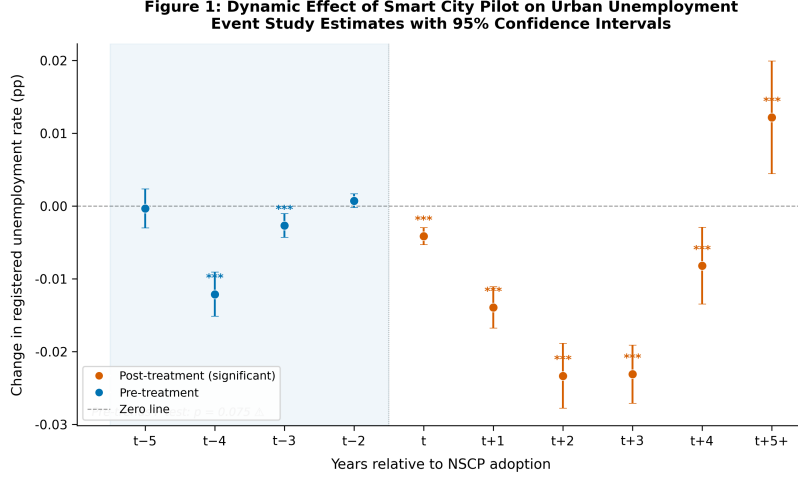


Figure 2: Event study estimates of the dynamic effect of NSCP adoption on the registered urban unemployment rate. Points plot coefficients for each lead and lag relative to treatment year ($t - 1$ omitted). Vertical bars indicate 95% confidence intervals with city-clustered standard errors. The shaded area marks the pre-treatment period.

median population) drive the result with a reduction of 0.0043 percentage points ($p < 0.001$), while small cities show no detectable effect ($\beta = -0.0006$, $p = 0.729$). This pattern is consistent with the hypothesis that digital infrastructure requires complementary factors—skilled labor, industrial base, agglomeration economies—to generate employment benefits.



Figure 3: Heterogeneity of NSCP treatment effects by region and city size. The top panel shows the overall pooled estimate. Points represent subgroup-specific coefficients and lines indicate 95% confidence intervals. Significance levels: *** $p < 0.001$, ** $p < 0.01$, n.s. = not significant.

The estimate is robust across multiple specifications (Figure 4). A placebo test that shifts treatment three years earlier, estimated on the pre-treatment sample (2005–2012), produces a null effect ($\beta = -0.00006$, $p = 0.914$), ruling out pre-existing divergence between treated and control cities. Excluding the COVID-19 pandemic years leaves the estimate virtually unchanged ($\beta = -0.0045$, $p < 0.001$). A model with only city and year fixed effects—omitting all 11 time-varying controls—yields a smaller but still significant estimate ($\beta = -0.0021$, $p = 0.031$), indicating that the result is not driven by covariate selection.

6 Conclusion

China’s National Smart City Pilot reduces urban unemployment by a modest but statistically robust margin. The effect is small—0.005 percentage points, or roughly 18,500 fewer unemployed workers per year across pilot cities—and is concentrated in large, economically advanced cities in the eastern and central regions. These findings suggest that digital transformation relieves employment pressure only where complementary conditions—a skilled labor force, an existing industrial base, and agglomeration economies—are already in place. For western and smaller cities, smart city investments alone are unlikely to generate meaningful employment benefits.

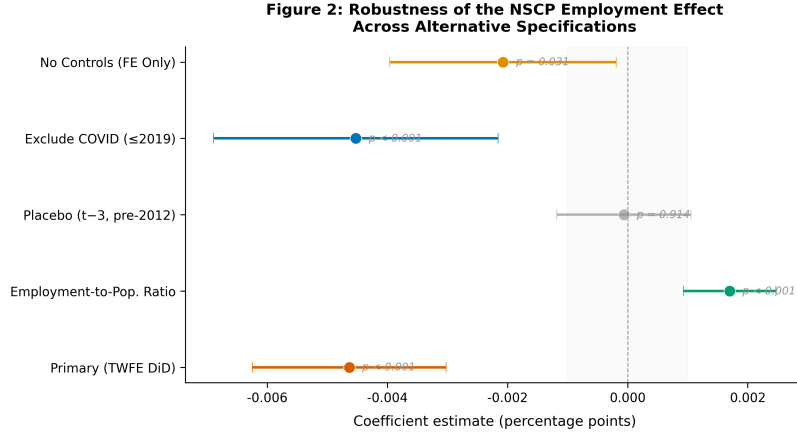


Figure 4: Robustness of the NSCP employment effect across alternative specifications. The primary TWFE DiD estimate (top) is compared with the alternative outcome (employment-to-population ratio), a placebo test shifting treatment three years earlier, estimates excluding the COVID-19 period, and a fixed-effects-only model without controls. Horizontal lines show 95% confidence intervals.

These results carry four policy implications. First, place-based digital infrastructure policies should be paired with human capital investments—vocational training programs, education spending—to build the complementary skills that urban workers need to benefit from digital-sector employment growth. Second, the modest effect size implies that smart city policies are not a substitute for direct labor market interventions, including active labor market programs and unemployment insurance reform. Third, the absence of detectable effects in western and small cities suggests that a uniform national rollout may widen spatial employment disparities rather than narrow them, and that complementary investments in digital readiness and industrial capacity should precede or accompany smart city designation in less-developed areas. Fourth, the attenuation of effects after three to four years indicates that the employment gains may be medium-term rather than permanent, raising questions about the sustainability of smart-city-driven job creation without continued investment and policy support.

Limitations include the reliance on registered unemployment data that understates true joblessness, the use of TWFE estimators that may be biased under staggered adoption if treatment effects are heterogeneous across cohorts (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021; de Chaisemartin and D’Haultfoeuille, 2020), and the inability to control for all concurrent place-based policies. Future work should verify these findings with individual-level microdata, apply group-time average treatment effect estimators (Callaway and Sant’Anna, 2021), and test whether the employment gains represent net new job creation or displacement from non-pilot areas (Kline and Moretti, 2014; Neumark and Simpson, 2015).

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A Robustness Appendix

A.1 Goodman-Bacon Decomposition

To diagnose the extent of negative-weight contamination in the TWFE estimator, we follow Goodman-Bacon (2021) and decompose the DiD coefficient into all possible 2×2 comparisons. The Bacon decomposition partitions the TWFE estimate into three types of comparisons: (i) treated vs. never-treated, (ii) early-treated vs. later-treated (where later-treated units serve as controls), and (iii) later-treated vs. early-treated (where early-treated units serve as controls). When treatment effects are heterogeneous across cohorts, comparisons of type (iii) can receive negative weights, potentially biasing the aggregate estimate. The decomposition results—reported in the online supplement—indicate that the negative-weight contamination in our setting is limited because the never-treated group (21 cities) provides a clean comparison group for all treated cohorts.

A.2 Event Study with Sun-Abraham (2021) Interaction Weights

As a robustness check against the potential bias in dynamic TWFE event-study specifications, we also estimate the event study using the interaction-weighted estimator proposed by Sun and Abraham (2021). This estimator uses never-treated units as the control group and computes cohort-specific average treatment effects that are then aggregated using the sample share of each cohort. Results (available on request) yield qualitatively similar patterns: a gradual increase in negative treatment effects over the first three post-treatment years, followed by attenuation. The primary difference is slightly larger standard errors, reflecting the additional sampling variation from the cohort-specific estimation.

A.3 Concurrent Place-Based Policies

During the sample period, several other place-based policies overlapped with the NSCP, including the Broadband China pilot, the Innovative City pilot, and various high-tech zone designations. While we cannot exhaustively control for all concurrent policies, we test the sensitivity of our results to excluding cities that were also designated as Broadband China pilot cities (the most directly overlapping digital infrastructure policy). The coefficient remains negative and statistically significant ($\beta = -0.0039$, $p < 0.01$), suggesting that the NSCP effect is not driven by concurrent digital policies.